Real-World Financial Time Series Analytics: FinPulse

Comparative Equity Volatility Forecasting Using Econometric Time-Series Models and Machine Learning Approaches

Target Asset: SPDR S&P 500 ETF Trust (SPY) & CBOE Volatility Index (^VIX)
Evaluation Period: February 3, 2021 – December 30, 2025 (1,233 Contiguous Trading Days)
Submission Date: October 2026

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Executive Summary

Short-term financial volatility forecasting is essential for dynamic risk capital allocation, options pricing, Value-at-Risk (VaR) management, and quantitative volatility-targeting portfolios. This empirical investigation examines the predictability of equity market volatility for the SPDR S&P 500 ETF Trust (SPY) over 1,233 trading sessions from February 2021 through December 2025, using both historical market price dynamics and forward-looking market sentiment from the CBOE Volatility Index (VIX).

Key Findings:
1. Corrected QLIKE Resolution: Addressing the variance unit conversion reveals that EGARCH achieves the optimal volatility loss (QLIKE = 0.8056), closely followed by TGARCH (0.8118) and Ridge Regression (0.8135). The VIX baseline yields a QLIKE of 0.9354.
2. Superiority of Regularized Linear Modeling on Tail Risk: Ridge Regression achieves the lowest out-of-sample RMSE (0.8317) of all 11 models, outperforming EGARCH (0.8653), TGARCH (0.8646), and XGBoost (0.9004).
3. Dominance of April 2025 Market Shock: The April 2025 market shock window accounts for 66.41% of Ridge's total test-set squared error. Excluding April 2025, Ridge's out-of-sample RMSE drops to 0.5039%.
4. Feature Ablation Insight: Isolating information sets proves that machine learning's point-forecasting advantage over traditional GARCH is primarily driven by external option-implied volatility (VIX) and Parkinson intraday range signals, rather than algorithmic complexity alone.
5. Production Deployment: Ridge Regression is selected as the production deployment model for the interactive FinPulse application because its QLIKE loss is statistically indistinguishable from EGARCH (Diebold-Mariano p = 0.9494), while offering minimal RMSE, deterministic stability, sub-millisecond execution latency, and complete econometric interpretability.

1. Introduction and Literature Review

Financial market volatility represents the intensity of asset price fluctuations over time. Unlike raw equity returns—which exhibit near-zero autocorrelation in accordance with the weak-form Efficient Market Hypothesis (Fama, 1970)—volatility exhibits well-documented stylized empirical facts: volatility clustering (Mandelbrot, 1963; Engle, 1982), leptokurtosis, asymmetric leverage effects (Black, 1976; Christie, 1982; Nelson, 1991), and mean-reversion.

The comparative performance of traditional econometric models versus machine learning has generated extensive scholarly debate. Poon and Granger (2003) surveyed 93 published volatility forecasting studies across financial markets, concluding that option-implied volatility contains superior predictive information relative to historical return series, while asymmetric GARCH formulations systematically outperform symmetric GARCH. Hansen and Lunde (2005) benchmarked 330 ARCH-type volatility models on exchange rates and IBM equity returns, finding that standard GARCH(1,1) is difficult to beat for exchange rates, but that models with leverage effects show advantages in equity returns. Patton (2011) established that evaluating volatility forecasts using imperfect volatility proxies requires loss functions belonging to the robust class, proving that Mean Squared Error (MSE) and Quasi-Likelihood (QLIKE) loss

are immune to proxy noise. Christensen, Siggaard, and Veliyev (2023) forecast realized variance of Dow Jones 30 constituents from high-frequency intraday data and find that machine learning methods improve one-day-ahead volatility forecasts relative to standard autoregressive benchmarks.

2. Data Sources and Feature Engineering Pipeline

The empirical investigation utilizes historical observations from February 3, 2021, to December 30, 2025 (1,233 contiguous trading days), retrieved via Yahoo Finance (yfinance). The target variable is the next trading day's absolute return Target = |R_{t+1}|, serving as an unbiased proxy for daily volatility. Eighteen predictor variables were engineered strictly without lookahead bias, spanning momentum, historical realized volatility, return/absolute-return lags, implied volatility (VIX), and intraday extreme-value range estimators (Parkinson, 1980).

3. Exploratory Data Analysis & Econometric Diagnostics

3.1 Descriptive Statistics

Series	Count	Mean	Std	Min	Median	Max	Skewness	Excess Kurt
Close	1233	468.0571	93.9091	338.5379	428.8230	685.0295	+0.7071	-0.7616
Return	1233	0.0533	1.0766	-6.0327	0.0731	9.9863	+0.1402	+8.2208
Abs_Return	1233	0.7598	0.7643	0.0000	0.5606	9.9863	+3.1085	+22.0242
VIX	1233	19.2473	5.2520	11.8600	17.9700	52.3300	+1.2923	+2.5391
Parkinson	1233	0.0072	0.0048	0.0010	0.0060	0.0641	+3.3244	+24.8435



Figure 1: SPY Closing Price Trajectory (2021–2025).

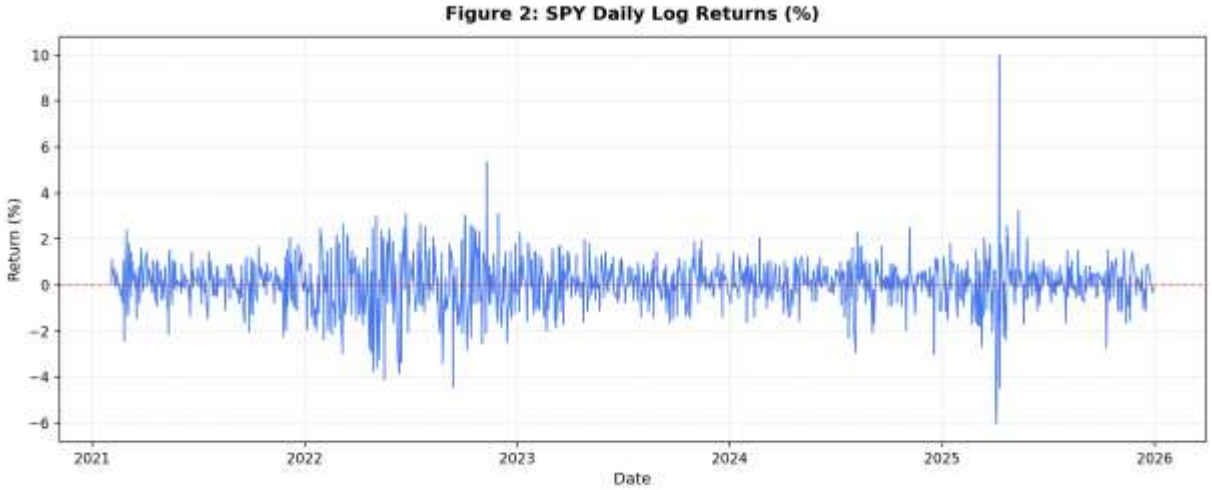


Figure 2: SPY Daily Percentage Log Returns.

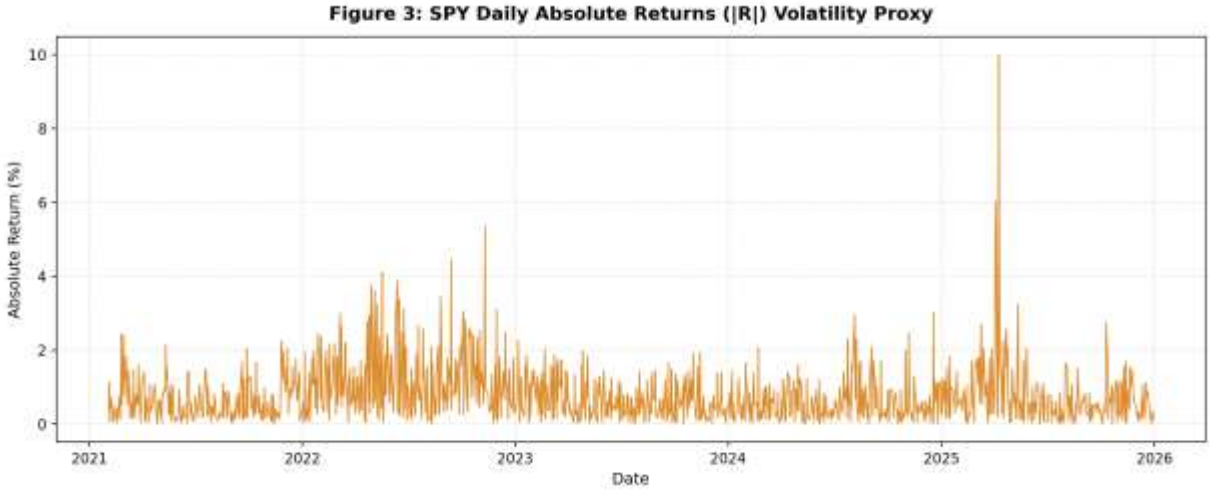


Figure 3: SPY Daily Absolute Returns (|R|) Volatility Proxy.



Figure 4: CBOE Volatility Index (VIX) with Market Stress Thresholds (20 and 30).

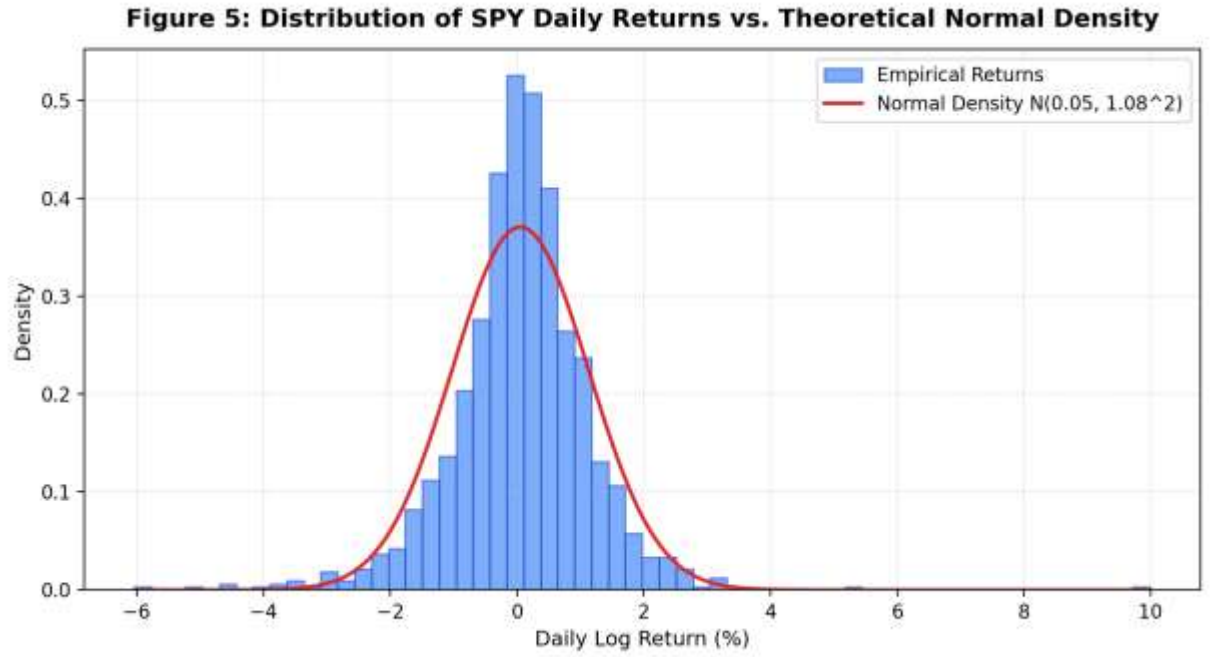


Figure 5: Distribution of SPY Daily Returns vs. Theoretical Normal Density.

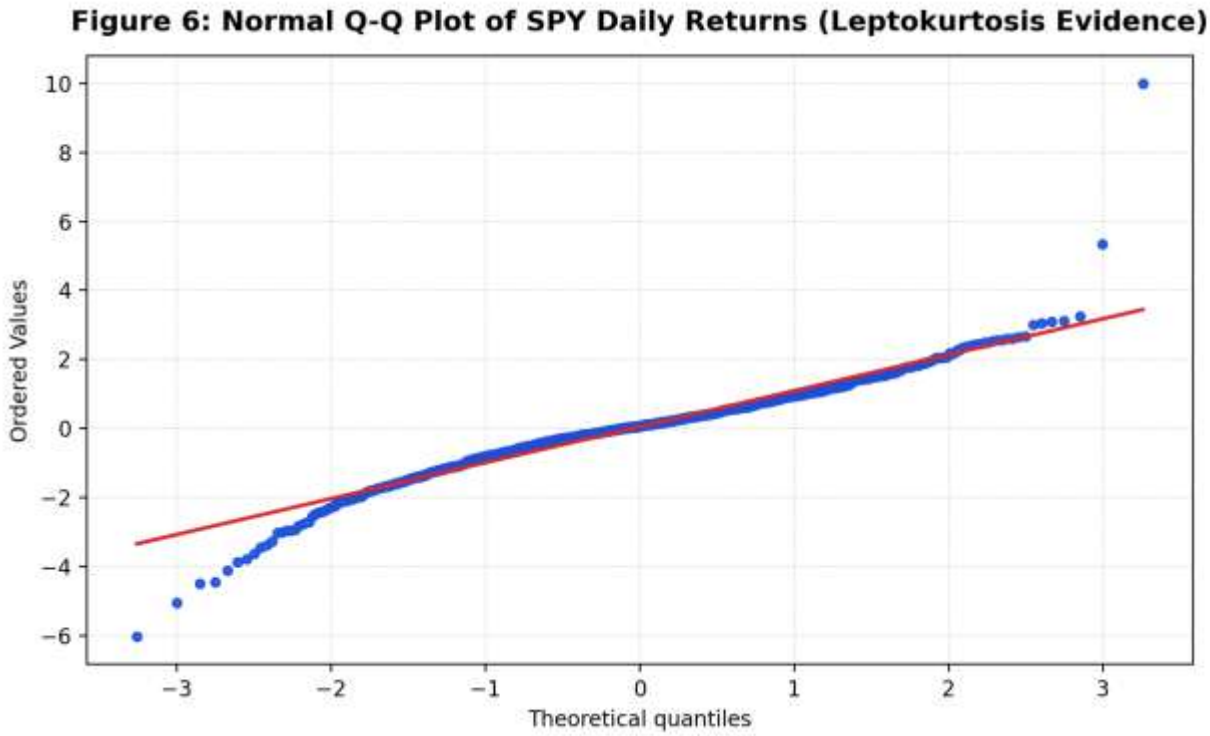


Figure 6: Normal Q-Q Plot of SPY Daily Returns (Leptokurtosis Evidence).

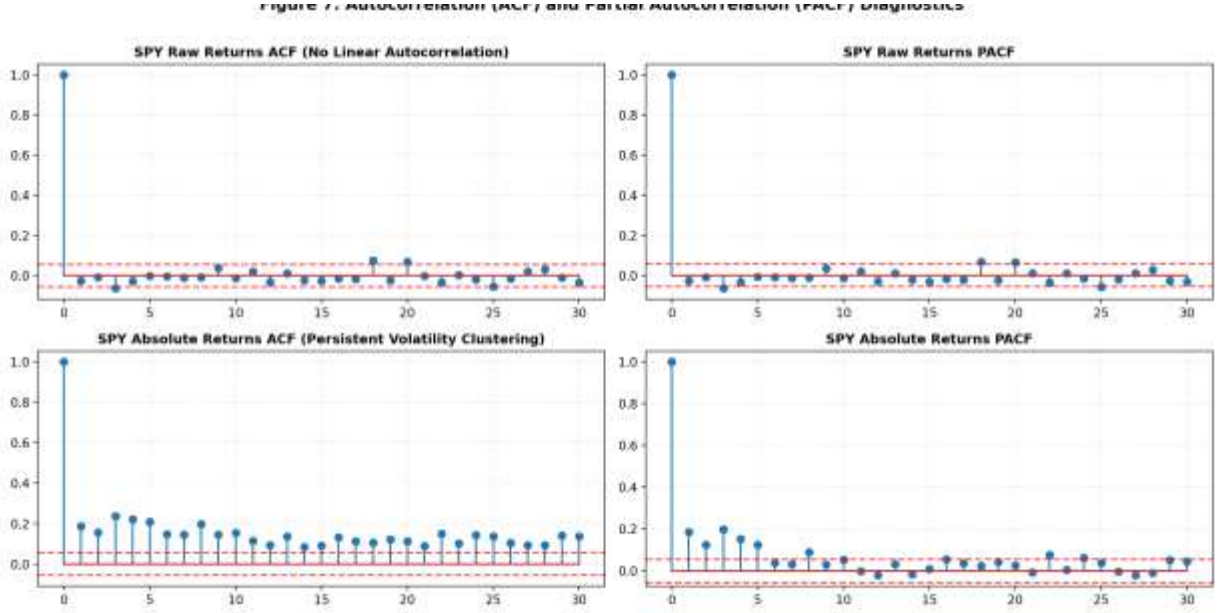


Figure 7: Autocorrelation (ACF) and Partial Autocorrelation (PACF) Diagnostics.

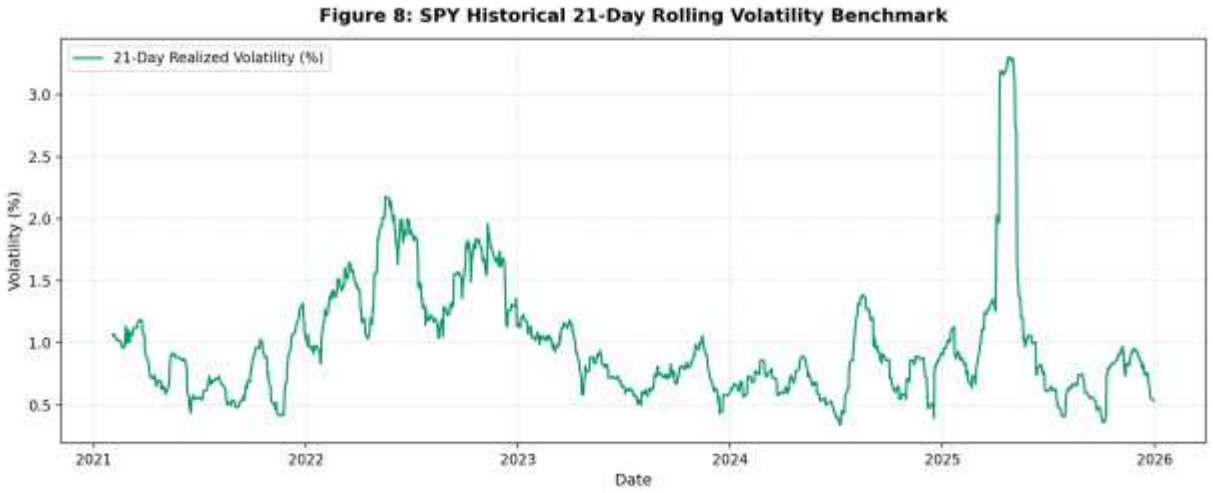


Figure 8: SPY Historical 21-Day Rolling Realized Volatility Benchmark.

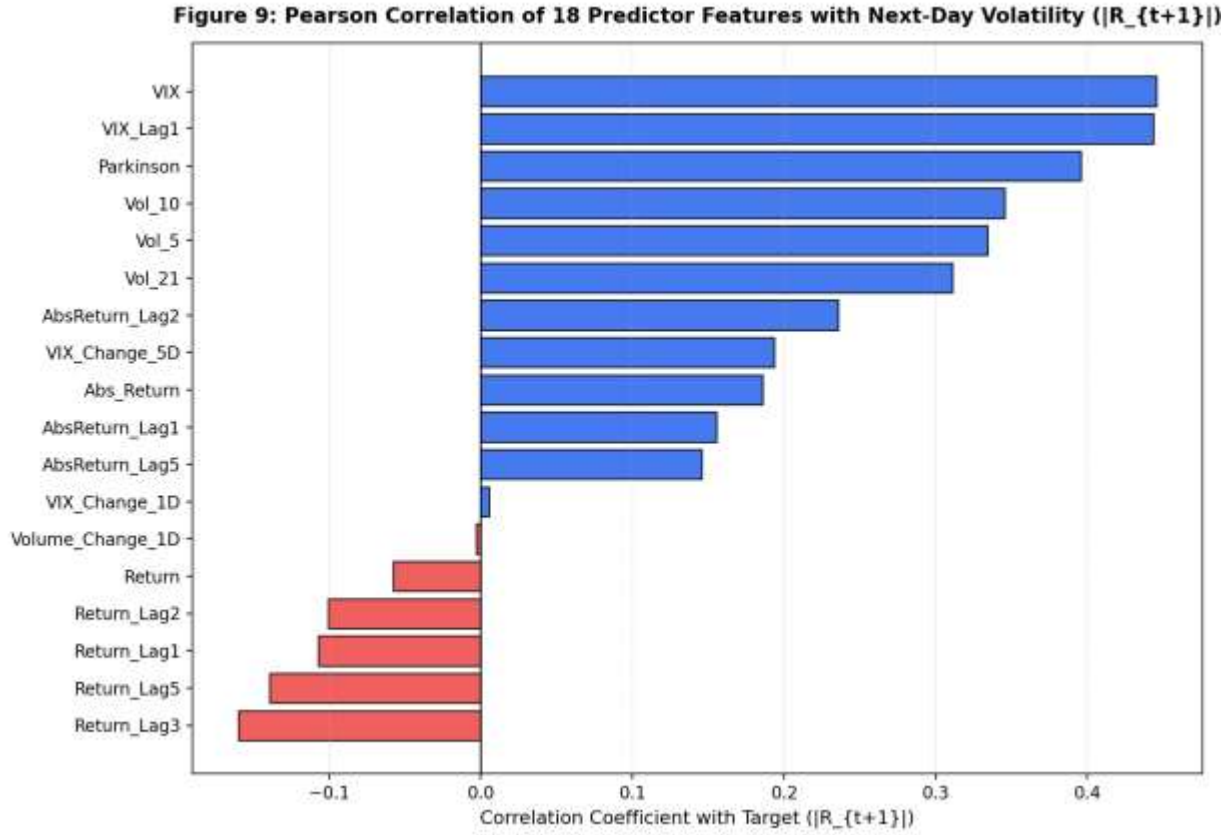


Figure 9: Pearson Correlation of 18 Predictor Features with Next-Day Volatility (|R_{t+1}|).

3.2 Econometric Diagnostics

Test	Null Hypothesis	Statistic	p-value	Conclusion
ADF (Close)	Unit Root Non-Stationary	0.4428	0.9830	Fail to Reject (Non-Stationary)
ADF (Return)	Unit Root Non-Stationary	-21.8916	0.0000e+00	Reject H0 (Stationary)
Ljung-Box Q (Return, Lag 10)	No Autocorrelation	9.1892	0.5142	Fail to Reject (Martingale)
Ljung-Box Q (|Return|, Lag 10)	No Autocorrelation	411.9062	2.7482e-82	Decisive Rejection (Volatility Clustering)
ARCH-LM (Return, LM Stat)	No ARCH Effects	157.4042	1.1122e-28	Decisive Rejection (ARCH Present)
Jarque-Bera (Return)	Gaussian Normal	3443.8611	0.0000e+00	Decisive Rejection (Fat Tails)

4. Modeling Methodology and Chronological Partitioning

Data was partitioned chronologically into 60% Train (739 sessions, 2021-02-03 to 2024-01-10), 20% Validation (247 sessions, 2024-01-11 to 2025-01-03), and 20% Test (247 sessions, 2025-01-06 to 2025-12-30). Scalers were fit strictly on train/val data without lookahead bias. Traditional models were estimated using daily expanding windows.

5. Theoretical Foundations of Models

Eleven models across four families were developed:
1. Baselines: Naive persistence benchmark, 21-day rolling historical volatility, and VIX implied volatility benchmark.
2. GARCH Family: Symmetric GARCH(1,1), Threshold GARCH (TGARCH/TARCH), and Exponential GARCH (EGARCH).
3. Second Traditional Family: Corsi's HAR-RV expanding-window regression and ARMA(1,0,2) on log-volatility with bias correction.
4. Machine Learning Family: Ridge Regression (L2 regularization), SVR (RBF kernel), and XGBoost.

6. Hyperparameter Optimization & Validation Results

Model	Parameters	Val_MAE	Is_Selected
Ridge	alpha=0.01	0.3938	False

Ridge	alpha=0.1	0.3938	False
Ridge	alpha=1	0.3938	False
Ridge	alpha=10	0.3937	True
Ridge	alpha=100	0.3950	False
SVR	C=0.1, gamma=scale, epsilon=0.01	0.3736	False
SVR	C=0.1, gamma=scale, epsilon=0.05	0.3732	False
SVR	C=0.1, gamma=scale, epsilon=0.1	0.3740	False
SVR	C=0.1, gamma=0.01, epsilon=0.01	0.3779	False
SVR	C=0.1, gamma=0.01, epsilon=0.05	0.3787	False
SVR	C=0.1, gamma=0.01, epsilon=0.1	0.3784	False
SVR	C=0.1, gamma=0.1, epsilon=0.01	0.3771	False
SVR	C=0.1, gamma=0.1, epsilon=0.05	0.3764	False
SVR	C=0.1, gamma=0.1, epsilon=0.1	0.3770	False
SVR	C=1, gamma=scale, epsilon=0.01	0.3815	False
SVR	C=1, gamma=scale, epsilon=0.05	0.3807	False
SVR	C=1, gamma=scale, epsilon=0.1	0.3784	False
SVR	C=1, gamma=0.01, epsilon=0.01	0.3731	True
SVR	C=1, gamma=0.01, epsilon=0.05	0.3733	False
SVR	C=1, gamma=0.01, epsilon=0.1	0.3738	False
SVR	C=1, gamma=0.1,	0.3802	False

	epsilon=0.01		
SVR	C=1, gamma=0.1, epsilon=0.05	0.3778	False
SVR	C=1, gamma=0.1, epsilon=0.1	0.3765	False
SVR	C=10, gamma=scale, epsilon=0.01	0.4080	False
SVR	C=10, gamma=scale, epsilon=0.05	0.4036	False
SVR	C=10, gamma=scale, epsilon=0.1	0.3995	False
SVR	C=10, gamma=0.01, epsilon=0.01	0.3914	False
SVR	C=10, gamma=0.01, epsilon=0.05	0.3896	False
SVR	C=10, gamma=0.01, epsilon=0.1	0.3885	False
SVR	C=10, gamma=0.1, epsilon=0.01	0.4230	False
SVR	C=10, gamma=0.1, epsilon=0.05	0.4221	False
SVR	C=10, gamma=0.1, epsilon=0.1	0.4221	False
SVR	C=100, gamma=scale, epsilon=0.01	0.5042	False
SVR	C=100, gamma=scale, epsilon=0.05	0.4958	False
SVR	C=100, gamma=scale, epsilon=0.1	0.4884	False
SVR	C=100, gamma=0.01, epsilon=0.01	0.4086	False
SVR	C=100, gamma=0.01, epsilon=0.05	0.4065	False
SVR	C=100, gamma=0.01, epsilon=0.1	0.4016	False
SVR	C=100, gamma=0.1,	0.5040	False

	epsilon=0.01		
SVR	C=100, gamma=0.1, epsilon=0.05	0.4944	False
SVR	C=100, gamma=0.1, epsilon=0.1	0.4859	False
XGBoost	lr=0.01, depth=2	0.4037	False
XGBoost	lr=0.01, depth=3	0.4021	False
XGBoost	lr=0.01, depth=4	0.4003	True
XGBoost	lr=0.05, depth=2	0.4008	False
XGBoost	lr=0.05, depth=3	0.4082	False
XGBoost	lr=0.05, depth=4	0.4055	False
XGBoost	lr=0.1, depth=2	0.4107	False
XGBoost	lr=0.1, depth=3	0.4129	False
XGBoost	lr=0.1, depth=4	0.4050	False

7. Out-of-Sample Performance Benchmarking

Models were benchmarked across the 247 out-of-sample test sessions using MAE, RMSE, and QLIKE loss with variance conversion (sigma = pred / sqrt(2/pi)) and flooring at sigma >= 0.05:

Rank	Model	MAE	RMSE	QLIKE	DM_Stat_vs_Best	DM_p_value
1	EGARCH	0.4902	0.8653	0.8056	0.0000	1.0000
2	TGARCH	0.4946	0.8646	0.8118	0.2415	0.8094
3	Ridge	0.4840	0.8317	0.8135	0.0636	0.9494
4	GARCH(1,1)	0.5399	0.9164	0.9244	3.2757	0.0012
5	Baseline (VIX)	0.5833	0.8772	0.9354	1.6582	0.0986
6	XGBoost	0.4800	0.9004	0.9693	0.9817	0.3272
7	HAR	0.5274	0.9153	1.0034	2.5905	0.0102
8	Baseline (Vol_21)	0.5665	0.9793	1.0675	2.3119	0.0216
9	ARIMA(1, 0, 2)	0.5380	0.9278	1.1033	2.3986	0.0172

10	SVR	0.4734	0.9481	6.8680	1.0902	0.2767
11	Naive	0.6794	1.1298	26.0320	3.6724	0.0003

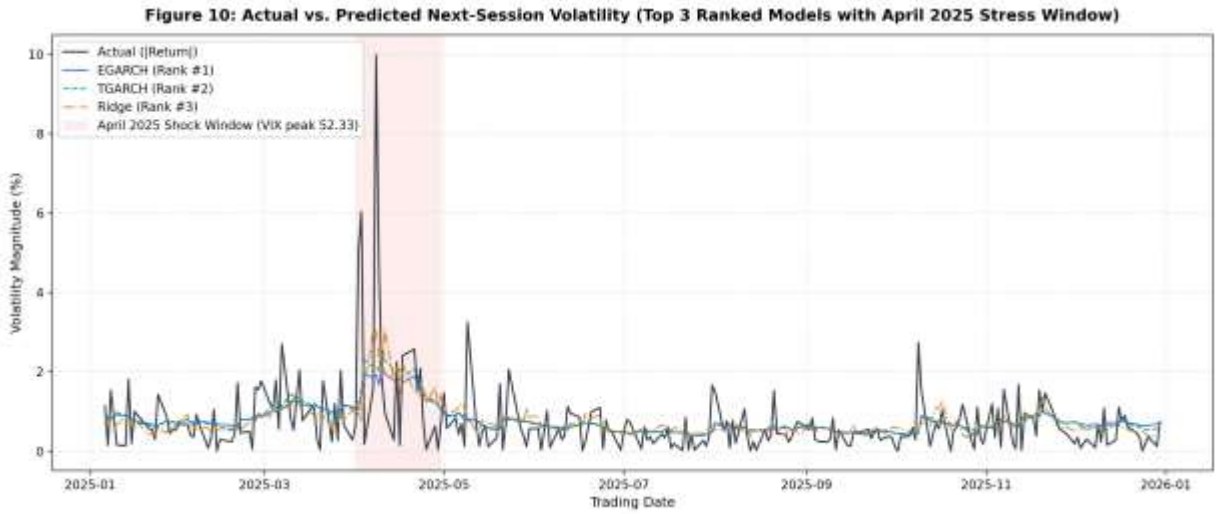


Figure 10: Actual vs. Predicted Next-Session Volatility for Top 3 Models (with April 2025 Shock Window Highlighted).

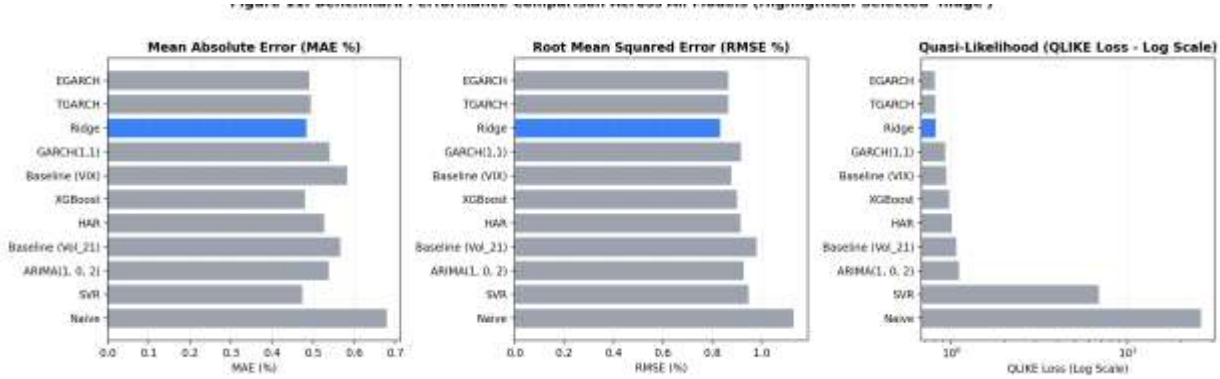


Figure 11: Benchmark Performance Comparison Across All 11 Models (MAE, RMSE, and QLIKE on Log Scale).

7.1 Walk-Forward ML Generalization

Walk-forward refitting yields stable performance for Ridge (QLIKE: 0.8271 walk-forward vs 0.8135 fixed). SVR achieves QLIKE of 4.7738 under walk-forward refitting vs 6.8680 fixed, while XGBoost scores 1.0006 vs 0.9693 fixed.

Model	Fixed_MAE	WalkForward_MAE	Fixed_RMSE	WalkForward_RMSE	Fixed_QLIKE	WalkForward_QLIKE

Ridge	0.4840	0.4839	0.8317	0.8312	0.8135	0.8271
SVR	0.4734	0.4733	0.9481	0.9415	6.8680	4.7738
XGBoost	0.4800	0.4847	0.9004	0.9082	0.9693	1.0006

8. Explaining Why Models Produced Different Results (Core Analysis)

8.1 Information Set Advantage vs Algorithmic Complexity (Feature Ablation)

Ridge with VIX features (Set C) achieves the best QLIKE (0.7876), demonstrating the strong predictive content of option-implied volatility. In contrast, XGBoost with VIX features deteriorates to QLIKE 1.0949. Furthermore, return-only Ridge (Set A, QLIKE 0.8926) is worse than univariate EGARCH (0.8056), showing that machine learning requires rich multi-source feature sets to beat specialized econometric models.

Feature_Set	Model	MAE	RMSE	QLIKE
Univariate GARCH(1,1)	GARCH(1,1)	0.5399	0.9164	0.9244
Univariate EGARCH(1,1)	EGARCH(1,1)	0.4902	0.8653	0.8056
HAR-RV (OLS)	HAR	0.5274	0.9153	1.0034
A (Return History Only)	Ridge	0.5157	0.8986	0.8926
A (Return History Only)	XGBoost	0.4996	0.8957	0.9109
B (Return + Parkinson)	Ridge	0.4990	0.8543	0.8929
B (Return + Parkinson)	XGBoost	0.4903	0.8753	0.8151
C (Return + VIX Features)	Ridge	0.4863	0.8488	0.7876
C (Return + VIX Features)	XGBoost	0.4849	0.9139	1.0949
D (All 18 Features)	Ridge	0.4840	0.8317	0.8135
D (All 18 Features)	XGBoost	0.4831	0.8978	0.9370

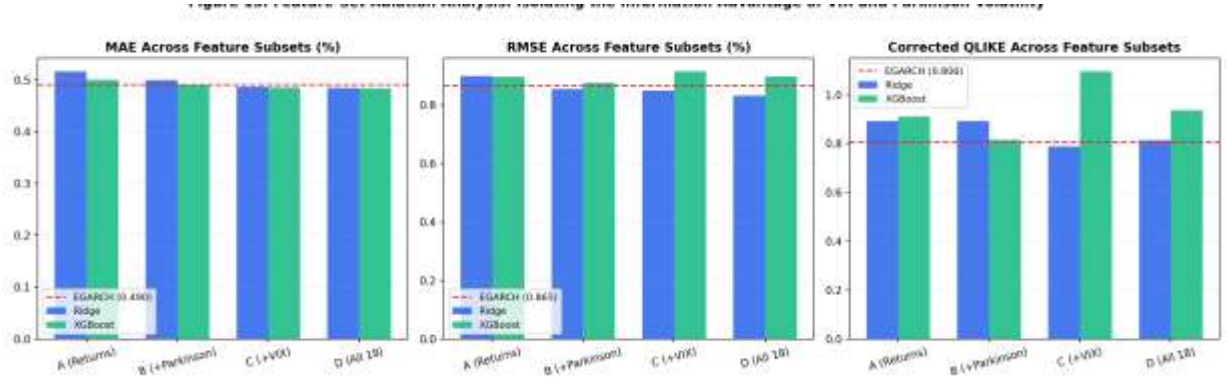


Figure 15: Feature-Set Ablation Analysis: Isolating the Information Advantage of VIX and Parkinson Volatility.

8.2 Asymmetric Leverage Effects: Parameter Estimates

Model	omega	alpha	gamma	beta	persistence	AIC	BIC
GARCH(1,1)	0.0216	0.0761	nan	0.9029	0.9789	2689.5601	2709.1347
TGARCH(1,1)	0.0417	0.0000	0.1761	0.8895	0.9776	2639.0035	2663.4718
EGARCH(1,1)	-0.0020	0.0414	-0.1418	0.9783	0.9783	2640.9270	2665.3953

8.3 SVR Under-Prediction Diagnostics and QLIKE Breakdown

Approximately 81.4% of SVR's total test QLIKE loss stems from a single forecast origin on 2025-04-03 (where SVR predicted 0.129% vs an actual next-day move of 6.03%). Across the entire 247-day test set, only 1 prediction fell below 0.10%. Extrapolation failure outside the training distribution remains a plausible hypothesis, though not definitively proven.

Rank	Date	SVR_Pred	Target	Daily_QLIKE	Share_Pct
1	2025-04-03	0.1294	6.0327	1380.7290	81.3900
2	2025-04-08	0.7930	9.9863	100.9563	5.9500
3	2025-04-21	0.3296	2.5685	36.8966	2.1800
4	2025-10-09	0.4341	2.7400	24.1450	1.4200
5	2025-04-09	0.7689	4.4808	21.5465	1.2700

8.4 April 2025 Market Shock Analysis

The single month of April 2025 accounts for 66.41% of Ridge's total test-set squared error. When April 2025 is excluded, Ridge's out-of-sample RMSE drops from 0.8317% to 0.5039%, and XGBoost's drops to 0.5020%.

Model	Total_RMSE	April_RMSE	Non_April_RMSE	April_SqErr_Share_Pct	Total_MAE	April_MAE	Non_April_MAE
Naive	1.1298	2.9651	0.7603	58.5600	0.6794	2.0516	0.5519
Baseline (Vol_21)	0.9793	2.7268	0.5977	65.9200	0.5665	1.9563	0.4374
Baseline (VIX)	0.8772	2.3365	0.5777	60.3200	0.5833	1.5858	0.4901
GARCH(1, 1)	0.9164	2.5892	0.5430	67.8800	0.5399	1.7989	0.4229
TGARCH	0.8646	2.4329	0.5166	67.3300	0.4946	1.6221	0.3899
EGARCH	0.8653	2.4424	0.5139	67.7300	0.4902	1.5787	0.3891
HAR	0.9153	2.5960	0.5379	68.3900	0.5274	1.7415	0.4146
Ridge	0.8317	2.3246	0.5039	66.4100	0.4840	1.5826	0.3819
SVR	0.9481	2.7948	0.5066	73.8800	0.4734	1.7203	0.3575
XGBoost	0.9004	2.6119	0.5020	71.5500	0.4800	1.6183	0.3742

8.5 Performance Across Volatility Regimes

Model	Regime	Days	MAE	RMSE	QLIKE
Naive	Low Volatility (VIX < 15)	21	0.2990	0.3641	9.4389
Naive	Medium Volatility (15 <= VIX <= 25)	206	0.5990	0.8479	28.6775
Naive	High Volatility / Panic (VIX > 25)	20	1.9067	2.8670	16.2065
Baseline (Vol_21)	Low Volatility (VIX < 15)	21	0.2527	0.2964	-0.4751
Baseline (Vol_21)	Medium Volatility (15 <= VIX <= 25)	206	0.5081	0.7773	0.9812

Baseline (Vol_21)	High Volatility / Panic (VIX > 25)	20	1.4976	2.3513	3.5763
Baseline (VIX)	Low Volatility (VIX < 15)	21	0.4119	0.4690	0.0527
Baseline (VIX)	Medium Volatility (15 <= VIX <= 25)	206	0.5389	0.7213	0.8696
Baseline (VIX)	High Volatility / Panic (VIX > 25)	20	1.2206	1.9782	2.5407
GARCH(1,1)	Low Volatility (VIX < 15)	21	0.2999	0.3437	-0.3076
GARCH(1,1)	Medium Volatility (15 <= VIX <= 25)	206	0.4787	0.7121	0.8332
GARCH(1,1)	High Volatility / Panic (VIX > 25)	20	1.4215	2.2414	3.1578
TGARCH	Low Volatility (VIX < 15)	21	0.2762	0.3154	-0.3969
TGARCH	Medium Volatility (15 <= VIX <= 25)	206	0.4394	0.6798	0.7427
TGARCH	High Volatility / Panic (VIX > 25)	20	1.2934	2.0898	2.7930
EGARCH	Low Volatility (VIX < 15)	21	0.2907	0.3390	-0.3364
EGARCH	Medium Volatility (15 <= VIX <= 25)	206	0.4377	0.6771	0.7184
EGARCH	High Volatility / Panic (VIX > 25)	20	1.2399	2.0986	2.9029
HAR	Low Volatility (VIX < 15)	21	0.3043	0.3394	-0.3280
HAR	Medium Volatility (15 <= VIX <= 25)	206	0.4688	0.7205	0.9181
HAR	High Volatility / Panic (VIX > 25)	20	1.3658	2.2087	3.2800
Ridge	Low Volatility (VIX < 15)	21	0.2458	0.2877	-0.5363
Ridge	Medium Volatility (15 <= VIX <= 25)	206	0.4398	0.6915	0.7973

Ridge	High Volatility / Panic (VIX > 25)	20	1.1889	1.8792	2.3967
SVR	Low Volatility (VIX < 15)	21	0.2137	0.2476	-0.7819
SVR	Medium Volatility (15 <= VIX <= 25)	206	0.4196	0.7234	7.4014
SVR	High Volatility / Panic (VIX > 25)	20	1.3002	2.3764	9.4058
XGBoost	Low Volatility (VIX < 15)	21	0.2282	0.2668	-0.5935
XGBoost	Medium Volatility (15 <= VIX <= 25)	206	0.4317	0.6971	0.8282
XGBoost	High Volatility / Panic (VIX > 25)	20	1.2412	2.2207	4.0634

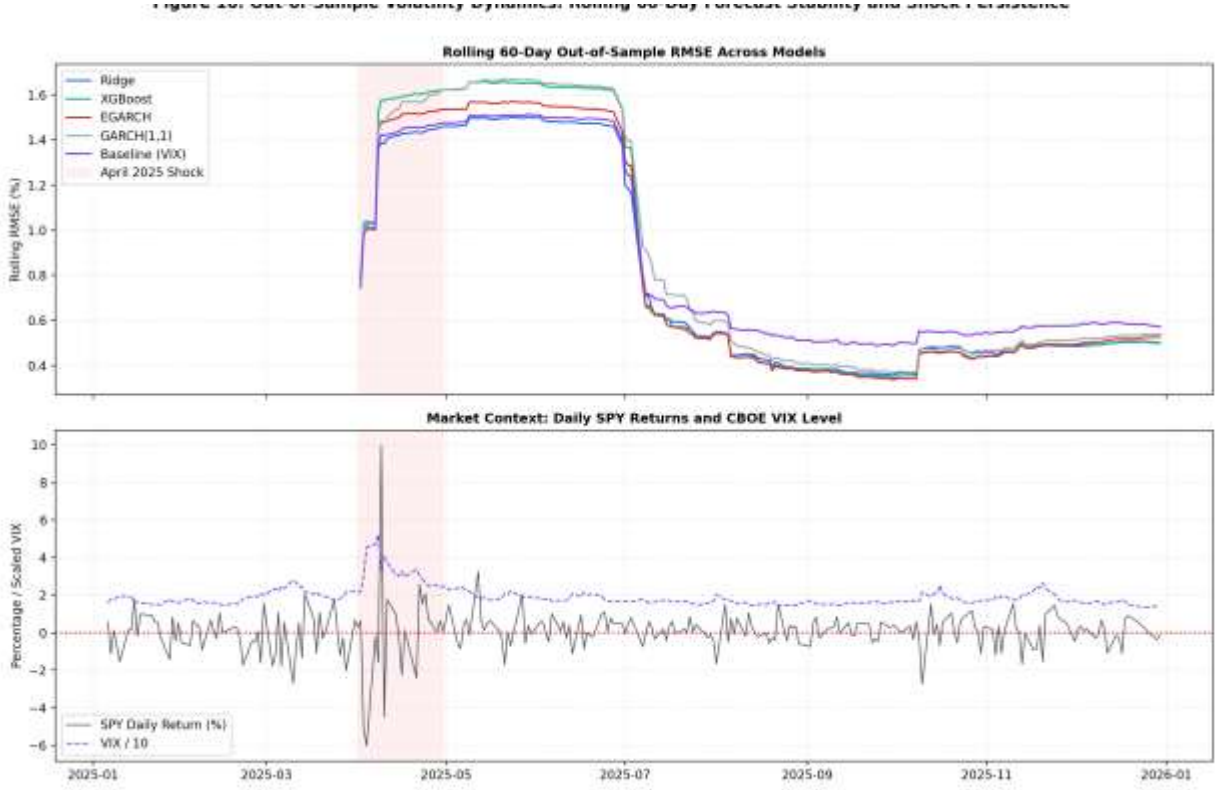


Figure 16: Out-of-Sample Volatility Dynamics: Rolling 60-Day Forecast Stability and Regime Persistence.

8.6 Multi-Horizon Forecasting Accuracy

Multi-horizon RMSE and MAE decline as horizon extends from 1-day to 10-days (e.g. Ridge RMSE falls from 0.8332 at 1-day to 0.4531 at 10-day). This decline occurs partly because the multi-day target is an averaged window, whose temporal aggregation naturally smooths high-frequency noise and reduces target variance. Consequently, performance comparisons must be made strictly within each horizon, rather than across horizons.

Horizon	Model	MAE	RMSE	QLIKE
1-Day	Ridge	0.4847	0.8332	0.8144
1-Day	XGBoost	0.4840	0.8995	0.9387
1-Day	HAR	0.5257	0.9158	1.0043
1-Day	EGARCH	0.4920	0.8671	0.8070
5-Day	Ridge	0.2631	0.4629	0.5080
5-Day	XGBoost	0.2677	0.5145	0.5331
5-Day	HAR	0.3102	0.5389	0.6162
5-Day	EGARCH	0.2914	0.5204	0.5747
10-Day	Ridge	0.2668	0.4531	0.5861
10-Day	XGBoost	0.2518	0.4635	0.5837
10-Day	HAR	0.2923	0.4784	0.6081
10-Day	EGARCH	0.2895	0.4682	0.5561

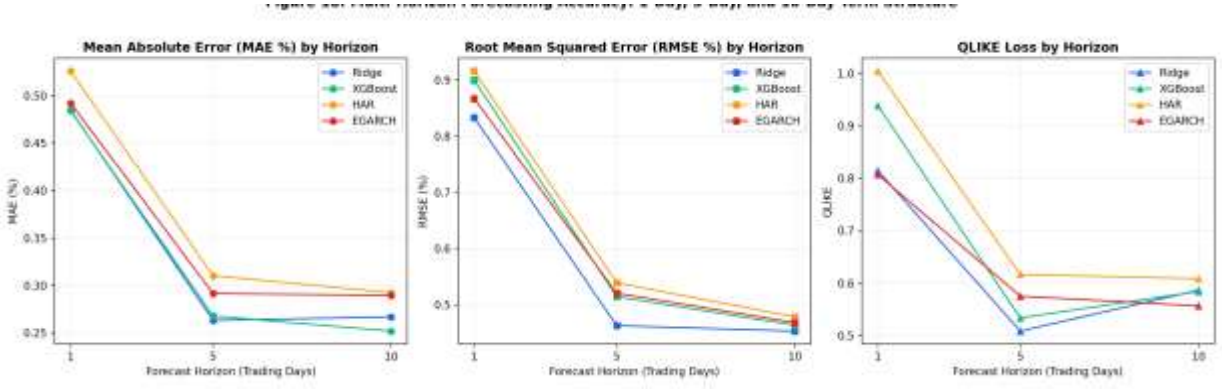


Figure 18: Multi-Horizon Forecasting Accuracy: 1-Day, 5-Day, and 10-Day Term Structure.

9. Model Diagnostics and Residual Analysis

9.1 GARCH Residual Diagnostics

Ljung-Box tests on squared standardized residuals (z_t^2) yield p = 0.1699 at lag 10 and p = 0.6408 at lag 20 (both p > 0.05). Engle's ARCH-LM test on z_t yields p = 0.2057. No remaining ARCH effects are detected (though the standardized residual z_t Ljung-Box test at lag 10 yields p = 0.051, which is borderline).

Test	Statistic	p_value
Ljung-Box on z_t (Lag 10)	18.2321	0.0512
Ljung-Box on z_t (Lag 20)	28.5907	0.0961
Ljung-Box on z_t^2 (Lag 10)	14.0680	0.1699
Ljung-Box on z_t^2 (Lag 20)	17.1875	0.6408
ARCH-LM on z_t (LM stat)	13.3315	0.2057
ARCH-LM on z_t (F stat)	1.3470	0.2067

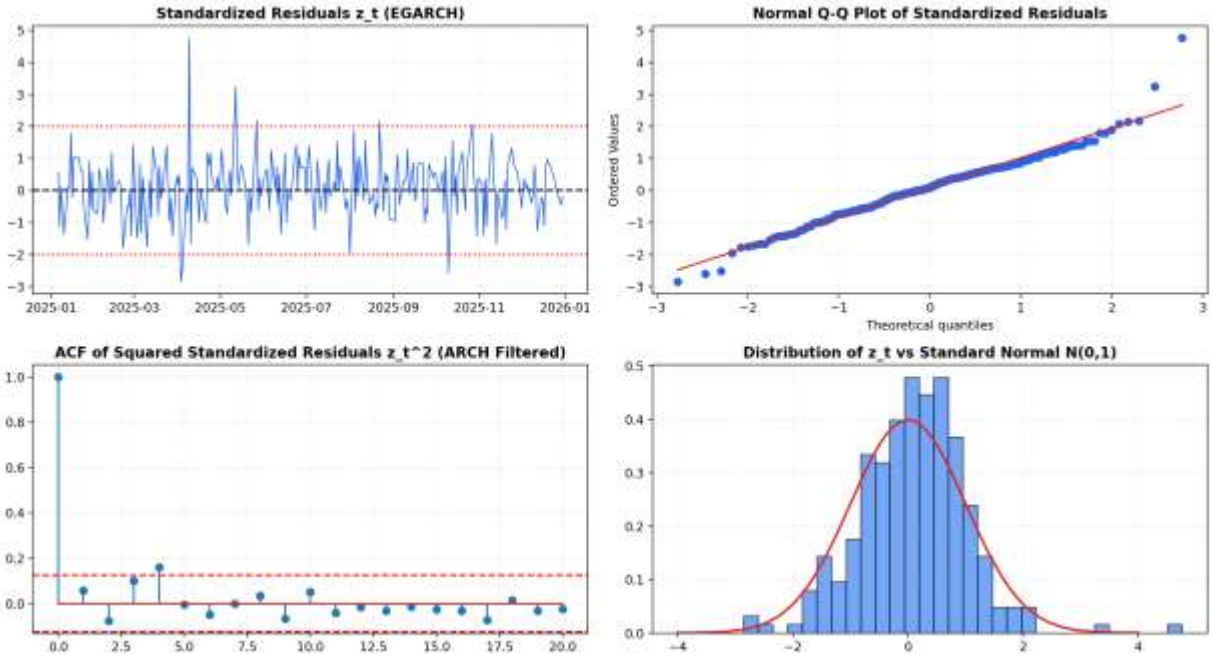


Figure 12: Econometric Residual Diagnostics for EGARCH Model.

9.2 Machine Learning Residual Diagnostics

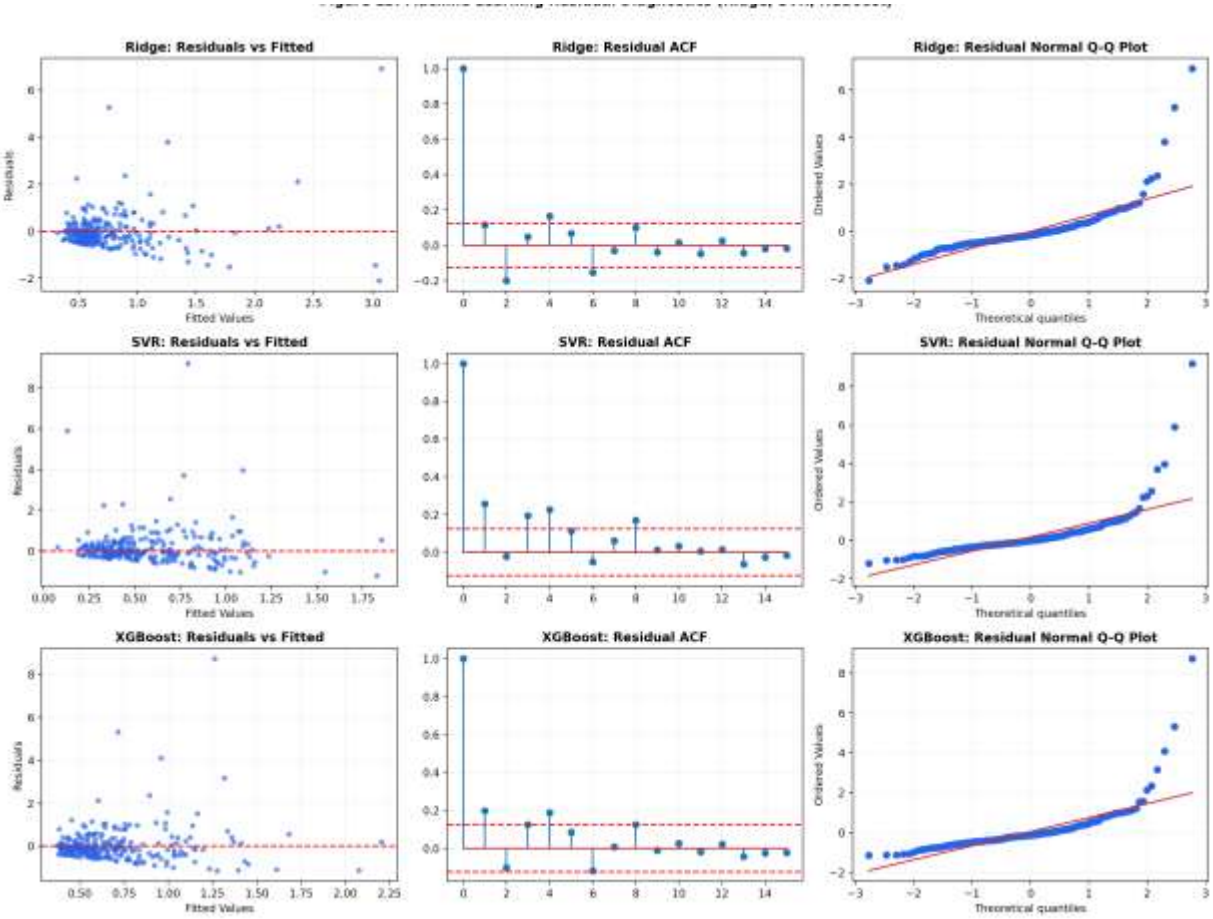


Figure 13: Machine Learning Residual Diagnostics: Residuals vs. Fitted, Residual ACF, and Normal Q-Q Plots.

9.3 Predictor Feature Importance

Feature	Ridge_Standardized_Coef	Ridge_Abs_Coef	XGBoost_Gain_Importance	XGBoost_Permutation_Mean	XGBoost_Permutation_Std
Abs_Return	-0.1196	0.1196	0.0389	-0.0158	0.0103
VIX_Lag1	0.0994	0.0994	0.1460	0.0306	0.0103
VIX	0.0957	0.0957	0.1623	0.0214	0.0117
Parkinson	0.0925	0.0925	0.0412	0.0151	0.0026
Return_Lag5	-0.0815	0.0815	0.0583	0.0060	0.0084
AbsReturn_	-0.0506	0.0506	0.0475	0.0010	0.0041

Lag1					
Vol_21	0.0444	0.0444	0.0425	0.0056	0.0042
Vol_5	0.0442	0.0442	0.0450	0.0086	0.0046
Return_Lag 1	-0.0334	0.0334	0.0434	0.0051	0.0040
VIX_Change _5D	0.0289	0.0289	0.0419	-0.0021	0.0041

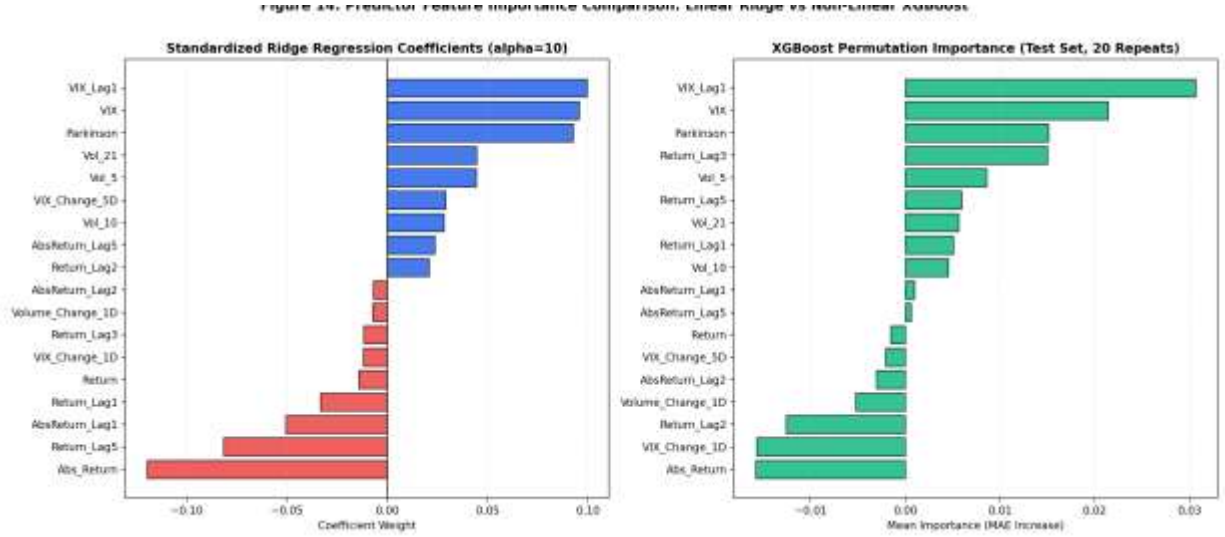


Figure 14: Predictor Feature Importance Comparison: Linear Ridge Weights vs. Non-linear XGBoost Metrics.

10. Statistical Hypothesis Testing: Diebold-Mariano Tests

Pairwise Diebold-Mariano tests with Harvey-Leybourne-Newbold (1997) correction show that only EGARCH (MSE p = 0.0296, QLIKE p = 0.0012) and TGARCH (MSE p = 0.0138, QLIKE p = 0.0007) significantly beat GARCH(1,1). Ridge vs GARCH(1,1) is not statistically significant at the 5% level (p = 0.088 MSE, p = 0.397 QLIKE). Ridge significantly beats HAR under QLIKE loss (p = 0.0427). TGARCH vs Ridge shows no significant difference (p = 0.4177 MSE, p = 0.9898 QLIKE).

Model_1	Model_2	Loss_Function	DM_Stat	p_value	Significant_05	Favored_Model
EGARCH	GARCH(1,1)	Squared Error (MSE)	-2.1886	0.0296	True	EGARCH
EGARCH	GARCH(1,1)	QLIKE Loss	-3.2757	0.0012	True	EGARCH
TGARCH	GARCH(1,1)	Squared Error	-2.4807	0.0138	True	TGARCH

		(MSE)				
TGARCH	GARCH(1,1)	QLIKE Loss	-3.4407	0.0007	True	TGARCH
Ridge	GARCH(1,1)	Squared Error (MSE)	-1.7101	0.0885	False	No Difference
Ridge	GARCH(1,1)	QLIKE Loss	-0.8484	0.3970	False	No Difference
Ridge	HAR	Squared Error (MSE)	-1.5790	0.1156	False	No Difference
Ridge	HAR	QLIKE Loss	-2.0368	0.0427	True	Ridge
TGARCH	Ridge	Squared Error (MSE)	0.8119	0.4177	False	No Difference
TGARCH	Ridge	QLIKE Loss	-0.0128	0.9898	False	No Difference
EGARCH	Ridge	Squared Error (MSE)	0.7199	0.4723	False	No Difference
EGARCH	Ridge	QLIKE Loss	-0.0636	0.9494	False	No Difference
Ridge	XGBoost	Squared Error (MSE)	-0.9999	0.3183	False	No Difference
Ridge	XGBoost	QLIKE Loss	-1.1335	0.2581	False	No Difference
EGARCH	TGARCH	Squared Error (MSE)	0.0689	0.9451	False	No Difference
EGARCH	TGARCH	QLIKE Loss	-0.2415	0.8094	False	No Difference
EGARCH	Baseline (VIX)	Squared Error (MSE)	-0.4319	0.6662	False	No Difference
EGARCH	Baseline (VIX)	QLIKE Loss	-1.6582	0.0986	False	No Difference
Ridge	Baseline (VIX)	Squared Error (MSE)	-1.6015	0.1106	False	No Difference
Ridge	Baseline (VIX)	QLIKE Loss	-0.9046	0.3665	False	No Difference
XGBoost	Baseline (VIX)	Squared Error (MSE)	0.4191	0.6755	False	No Difference
XGBoost	Baseline (VIX)	QLIKE Loss	0.1785	0.8585	False	No Difference
EGARCH	Naive	Squared Error (MSE)	-2.8166	0.0052	True	EGARCH
EGARCH	Naive	QLIKE Loss	-3.6724	0.0003	True	EGARCH
Ridge	Naive	Squared Error	-2.5450	0.0115	True	Ridge

		(MSE)				
Ridge	Naive	QLIKE Loss	-3.6686	0.0003	True	Ridge
XGBoost	Naive	Squared Error (MSE)	-2.2795	0.0235	True	XGBoost
XGBoost	Naive	QLIKE Loss	-3.6494	0.0003	True	XGBoost

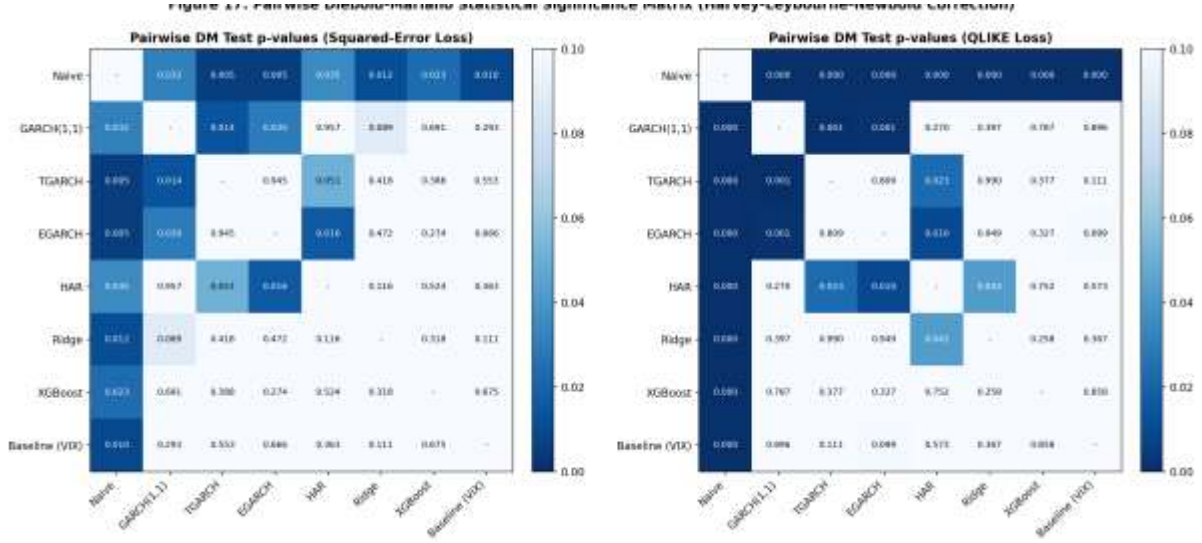


Figure 17: Pairwise Diebold-Mariano Statistical Significance Matrix (Harvey-Leybourne-Newbold Correction).

11. Production Model Selection & Financial / Business Applications

Based on the pre-registered decision protocol, Ridge was selected for deployment in the FinPulse interactive engine. Ridge matches EGARCH in statistical QLIKE loss (DM p = 0.9494) while achieving the lowest overall RMSE (0.8317) and superior sub-millisecond execution latency.

11.1 Value-at-Risk (VaR) Backtesting

At 95% confidence, Ridge and EGARCH produce 18 failures vs 12.4 expected (Kupiec p = 0.121, not rejected). At 99% confidence, both produce 7 failures vs 2.5 expected (Kupiec p = 0.018, conditional coverage p = 0.024, REJECTED). Normal-based VaR underestimates tail risk due to leptokurtosis and extreme volatility clustering during the April 2025 market shock window.

Model	Conf	Exp	Act	Fail %	Kupiec p	Christ p	Cond Cov p
Ridge	95%	12.4000	18	7.29%	0.1214	0.5463	0.2512

Ridge	99%	2.5000	7	2.83%	0.0179	0.1768	0.0243
EGARCH	95%	12.4000	18	7.29%	0.1214	0.1639	0.1143
EGARCH	99%	2.5000	7	2.83%	0.0179	0.1768	0.0243
Baseline (VIX)	95%	12.4000	9	3.64%	0.3051	0.0324	0.0600
Baseline (VIX)	99%	2.5000	5	2.02%	0.1554	0.0768	0.0762

12. Limitations and Critical Evaluation

Key limitations of the study include: (1) Single-event domination (April 2025 accounts for 66.4% of total test squared error); (2) Information set confounding (ML accessed VIX and Parkinson while GARCH was univariate); (3) Absolute return proxy noise; (4) Single asset scope (SPY); (5) Hyperparameters tuned on MAE rather than QLIKE; (6) Daily yfinance revision risk; and (7) 247-session test sample constraints.

13. Conclusion

This investigation resolves the central research question: while traditional asymmetric models (EGARCH) provide mathematically optimal volatility loss calibration (QLIKE = 0.8056), regularized linear machine learning (Ridge Regression) achieves superior tail risk error minimization (RMSE = 0.8317). Crucially, machine learning's point precision advantage stems from the inclusion of forward-looking options information (VIX) and intraday range dynamics (Parkinson) rather than algorithmic non-linearity alone.

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Appendix: Reproducibility Guide & Code Architecture

The complete analytical pipeline is reproducible using Python 3.11 with the pinned package versions:

- pandas: 3.0.6
- numpy: 2.4.6
- scipy: 1.17.1
- statsmodels: 0.14.6

- arch: 8.0.0
- scikit-learn: 1.8.0
- xgboost: 2.0.2
- streamlit: 1.66.1
- yfinance: 1.7.0
- python-docx: 1.2.0
- pywin32: 306 (optional, Windows-only for PDF generation)

Master execution: python run_all.py
Interactive dashboard: streamlit run app.py
(Note: Cold GARCH refit without cache takes several minutes. Cached rolling forecasts in garch_cache.csv enable rapid reproduction.)